

Software Architecture Justification Document

Course: Software Engineering Lab (Lab 3)

Problem Statement #33: Micro-Lending & Peer Credit Risk Assessor

Architecture Selection

We chose **Layered (N-Tier) Architecture** for the **Micro-Lending & Peer Credit Risk Assessor System**.

Architectural Choice & Scenario-Related Reasons

- **Deterministic Separation of Financial Workflows:**

Micro-lending platforms require strict functional boundaries between client presentation, regulatory business rules, and persistent storage. By organizing the system into Presentation, Business Logic, and Data Persistence layers, core heuristic evaluation models (`Credit Risk Engine`) and repayment schedule generation (`Repayment & Schedule Component`) are decoupled from frontend UI changes. This ensures heuristic credit scoring algorithms can be updated or calibrated without disrupting active client sessions or database access routines.

- **Simplicity in Preserving ACID Transaction Boundaries:**

A primary non-functional requirement (NFR-001) dictates that loan repayments and EMI disbursements execute strictly within database ACID transactions to prevent double-spending. While distributed architectures (such as microservices) introduce substantial data consistency challenges across network partitions, a layered architecture centralizes business execution against a unified persistence component (`Database Access Component`). This enables atomic, ledger-level transaction rollbacks during high-concurrency fund distributions without the operational overhead of distributed two-phase commit protocols.

Security Advantage

- **Isolation of Sensitive Financial & Personal Data:**

Layered architecture enforces strict linear access: client-facing components cannot interact directly with the database. The intermediate business tier acts as an isolation barrier where authentication and access policies are strictly evaluated before data retrieval. Furthermore, the persistence layer can centrally govern AES-256 at-rest encryption and enforce TLS 1.2+ transit rules (NFR-002), preventing sensitive financial profile records (debt-to-income ratios, bank details, credit scores) from ever being exposed to untrusted external client requests.

Performance Benefit

- **Low-Latency In-Memory Execution:**

Unlike microservices, where internal component-to-component coordination suffers from network hop latency and JSON/gRPC serialization overhead, components within the Business Logic Layer communicate via low-overhead in-memory method invocations. This enables rapid execution of complex loan-matching filters, automated schedule calculations, and heuristic risk scoring within the mandated sub-2-second response window (FR-003) under high simulated load.